

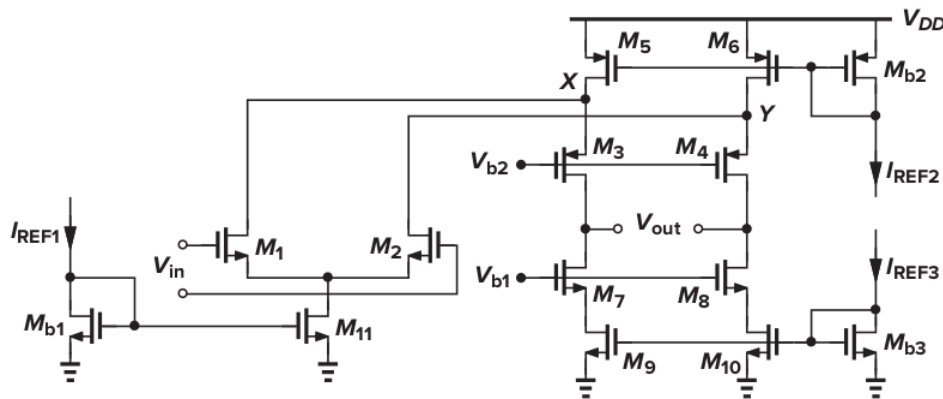
Name :

ID. No.

Q 1: For the circuit shown below, fill the given table (values with proper sign and units). The laser diode has resistance R_L . Include the loading effects of feedback network. For all transistors $g_m = 1 \text{ mS}$, $R_D = R_F = R_L = R_M = 10 \text{ k}\Omega$, $r_{o1} = \infty$, $r_{o2} = 100 \text{ k}\Omega$.

	Feedback factor	-0.5 A/A
	Open loop input impedance	952.38Ω
	Open loop output impedance seen by R_L	$105 \text{ k}\Omega$
	Open loop gain	-8.286 A/A
	Closed loop input impedance	185.18Ω
	Closed loop output impedance seen by R_L	$540.02 \text{ k}\Omega$
	Closed loop gain	-1.611 A/A

Q 2: For the circuit shown below, $|V_{ov}| = 0.15 \text{ V}$, $r_o = 50 \text{ k}\Omega$, $g_m = 1 \text{ mS}$, $|V_{th}| = 0.3 \text{ V}$ of all transistors, $V_{DD} = 3 \text{ V}$. V_{b2} is set to its maximum value and V_{b1} to its minimum value.



$V_{in,cm,min}$	0.6 V	$V_{b1,min}$	0.6 V
$V_{in,cm,max}$	$3 \text{ V} \text{ (3.15 V)}$	$V_{b2,max}$	2.4 V
Maximum output voltage swing	4.8 V	Voltage gain	833.33 V/V

What will be values in table below if $V_{b2} = V_{b2,max} - 0.1 \text{ V}$ and $V_{b1} = V_{b1,min} + 0.1 \text{ V}$?

$V_{in,cm,max}$	$3 \text{ V} \text{ (3.05 V)}$
Maximum output voltage swing	4.4 V